

MT IDA, AR, USA

WMO: 723435

Lat: 34.547N

Lon: 93.578W

Elev: 702

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.5	(c) 22.1	(d) 6.4	(e) 7.8	(f) 27.9	(g) 10.0	(h) 9.4	(i) 30.8	(j) 15.6	(k) 45.0	(l) 12.9	(m) 49.0	(n) 1.4	(o) 320	(p) 0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.1	(c) 97.7	(d) 74.9	(e) 95.0	(f) 75.2	(g) 92.3	(h) 74.9	(i) 78.9	(j) 89.4	(k) 78.0	(l) 88.8	(m) 77.1	(n) 87.8	(o) 4.5	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	140.6	82.9	75.1	135.1	82.2	74.0	130.1	81.4	42.9	89.7	41.8	89.1	40.8	87.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.2	(b) 10.6	(c) 9.3	(e) 9.8	(f) 101.3	(g) 4.0	(h) 4.1	(i) 7.0	(j) 104.3	(k) 4.7	(l) 106.7	(m) 2.4	(n) 109.0	(o) -0.5	(p) 112.0		
(a) 12.2	(b) 10.6	(c) 9.3	(e) 9.7	(f) 81.0	(g) 4.2	(h) 1.9	(i) 6.7	(j) 82.4	(k) 4.2	(l) 83.4	(m) 1.8	(n) 84.5	(o) -1.2	(p) 85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	41.7	44.9	53.3	61.5	69.0	77.2	80.4	80.3	73.2	61.9	51.5	43.8		
	DBStd	15.99	10.33	10.36	9.46	7.95	6.68	4.42	4.49	4.78	6.53	8.70	9.34	9.46		
	HDD50	935	300	207	73	9	1	0	0	0	0	10	92	243		
	HDD65	3117	725	566	379	157	40	0	0	0	11	165	415	659		
	CDD50	5187	42	64	174	355	589	816	943	940	697	379	137	51		
	CDD65	1893	2	2	15	52	163	366	478	475	259	69	10	2		
	CDH74	19165	5	9	96	421	1340	3697	5124	5170	2605	642	53	3		
	CDH80	8518	0	0	8	53	327	1569	2576	2685	1125	170	5	0		
(14) Wind	WSAvg	2.8	3.3	3.4	3.8	3.9	2.9	2.3	1.8	1.9	1.8	2.3	2.8	3.0		
(15) Precipitation	PrecAvg	54.3	4.4	3.9	5.0	5.2	6.1	4.2	3.7	2.9	4.5	4.8	4.6	5.0		
	PrecMax	76.4	8.1	8.4	11.5	11.1	15.1	8.5	8.7	11.1	12.7	11.4	11.6	8.1		
	PrecMin	33.5	0.9	1.7	1.5	2.1	2.0	0.8	0.8	0.6	0.0	0.0	1.1	0.4		
	PrecStd	10.4	1.9	1.8	2.2	2.1	3.0	2.3	2.0	2.1	2.7	3.1	3.0	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.8	75.2	81.0	84.3	89.3	96.7	100.4	102.0	97.1	88.9	80.1	73.1		
		MCWB	60.2	60.4	63.4	66.6	72.0	74.6	75.1	75.6	73.3	69.9	66.4	63.7		
	2%	DB	68.8	70.9	76.7	81.3	86.2	92.9	97.3	98.9	93.1	84.4	74.7	69.5		
		MCWB	59.3	61.7	61.8	65.7	71.1	75.1	75.8	74.9	73.2	69.7	63.5	63.4		
	5%	DB	65.0	67.2	73.0	79.0	83.7	90.5	94.2	95.5	90.1	81.1	71.8	65.4		
		MCWB	57.8	58.8	60.4	64.7	70.6	74.5	76.0	75.3	72.3	67.5	62.7	58.6		
	10%	DB	61.0	63.5	70.1	76.0	81.3	88.2	91.3	92.4	86.9	77.1	68.0	61.1		
		MCWB	52.0	55.6	59.3	63.7	69.8	74.1	75.9	74.9	72.0	66.0	60.0	53.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.4	66.2	67.0	71.5	76.0	79.3	80.3	80.5	77.8	74.8	69.9	66.6		
		MCDB	71.9	71.2	75.5	79.6	84.4	89.8	89.7	89.9	87.3	82.9	75.1	70.9		
	2%	WB	62.9	64.0	65.2	69.4	74.3	77.8	78.8	79.0	76.3	72.7	67.0	63.9		
		MCDB	66.9	68.4	72.8	77.3	82.2	88.1	89.8	89.2	86.1	80.1	72.7	67.7		
	5%	WB	59.1	60.6	63.3	68.0	73.0	76.6	78.0	77.9	75.1	70.8	64.2	59.7		
		MCDB	63.6	65.7	70.5	75.7	80.7	86.3	89.3	88.7	84.7	77.2	69.3	64.1		
	10%	WB	52.6	56.9	60.9	66.4	71.4	75.5	77.1	76.9	73.9	68.5	61.2	54.9		
		MCDB	58.0	61.8	68.6	73.5	79.0	84.6	88.1	87.5	83.1	74.7	66.6	59.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	20.4	22.3	22.3	20.0	20.5	21.1	22.4	24.2	23.7	22.5	21.0		
		MCDBR	25.8	24.7	24.7	24.4	22.6	23.7	24.6	27.2	27.3	26.6	24.4	25.1		
		MCWBR	17.7	17.0	13.6	12.0	10.3	8.8	7.2	8.2	9.5	12.8	14.6	17.9		
		MCDBR	19.6	20.1	20.3	17.8	17.7	19.7	21.1	21.6	21.7	19.7	18.8	20.7		
	5% WB	MCWBR	17.0	16.1	13.0	10.8	9.6	8.5	7.8	8.0	8.6	11.0	14.3	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.324	0.355	0.387	0.419	0.459	0.471	0.469	0.399	0.355	0.332	0.311		
	taud		2.496	2.488	2.426	2.339	2.310	2.225	2.229	2.218	2.387	2.489	2.508	2.536		
	Ebn at Noon		281	293	291	285	276	264	260	258	273	279	275	279		
	Edh at Noon		27	30	35	40	41	45	44	44	36	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		804	933	1298	1623	1764	1989	1958	1808	1551	1197	854	697		
	RadStd		72	133	98	95	164	185	122	137	129	170	102	85		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page